Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2423)

# Key

## SECTION – I

1. D
2. B
3. B
4. D
5. A
6. D
7. A
8. C
9. D
10. C
11. D
12. B
13. C
14. B
15. A
16. C
17. 4132
18. B
19. D
20. 3142
21. 3421
22. B
23. C
24. A

## SECTION – II

1. D
2. B
3. D
4. C
5. B
6. 52
7. C
8. A
9. 52
10. C
11. 3
12. 2
13. C
14. D
15. A
16. B
17. A
18. 66
19. C
20. 50

## SECTION – III

1. C
2. 76
3. B
4. 8
5. 8
6. B
7. A
8. C
9. C
10. 3
11. 1
12. D
13. D
14. 30
15. D
16. A
17. A
18. 65
19. B
20. C
21. C
22. 40

# Solutions

## SECTION – I

**Solutions for questions 1 to 4:**

**Number of words and Explanatory notes for RC:**

Number of words: 508

**1.** Option A: Choice A has been mentioned as a fact in the passage. It is not the author's main viewpoint. Refer to para 2: These government laboratories provide a bridge between the public and private sectors. Sometimes governments simply copy what private firms are doing. ....... Refer to para 3: The units also provide a connection with academia. Hence, choice A is not the answer.
Option B: Choice B is partly true as evident from the lines in the penultimate para of the passage: What difference can a few reformers in mayor's offices in Boston or New Orleans make? Bureaucracies are designed to kill innovation in the name of predictability. And a change in the political wind can reverse sensible changes. But choice B does not correctly capture the author's viewpoint in the passage. Hence, choice B is a generalization that cannot be inferred from the passage. It is not the author's viewpoint.
Option C: Choice C may be true elsewhere but the word 'only' makes it narrow in scope. Hence, choice C is not the answer.
Option D: Refer to the last para. While showing some let downs of having innovation centres linked up with governments, the author is still happy that there is progress, however small it is. .... Still, something is better than nothing. And the various innovation centres have a growing number of somethings to their name. Reforming government is hard and often boring work. The innovation labs are making it a bit faster and a lot more interesting. Hence, choice D is the answer. Choice (D)

**2.** Option A: A narrative passage tells a story, usually from one person's viewpoint. A narrative passage is a sequence of events and follows a story telling format. A narrative passage neither presents any analysis nor evokes any emotions. This passage is not narrative. Hence, choice A is not the answer.
Option B: Analysis involves examining aspects of a situation in its plusses and minuses, and making an evaluation at the end of it. In this passage, the author analyzes a situation and weighs it up. The purpose of this passage is to offer an analysis: Still, something is better than nothing. And the various innovation centres have a growing number of somethings to their name. Hence, choice B is the answer.
Option C: A descriptive passage makes a discussion vivid with detail. Here the author is not using a descriptive tone. Hence, choice C is incorrect.
Option D: The passage is not argumentative. There is no debate, i.e., the passage does not present arguments and counterarguments for any idea or concept. The author refrains from delivering an argument; neither does he try to convince people of an argument. Overall, the style of the passage is analytical. Hence, choice D is incorrect. Choice (B)

© Advanced Educational Activities Pvt. Ltd. 95B, Siddamsetty Complex, Park Lane, Secunderabad – 500003.
All rights reserved. No part of this material may be reproduced, in any form or by any means, without permission in writing.
This course material is only for the use of bonafide students of Advanced Educational Activities Pvt. Ltd (T.I.M.E.), and its licensees/franchisees and is not for sale. (13 pages) (ageg/ageh)

**3.** Refer to the penultimate para. Bureaucracies are designed to kill innovation in the name of predictability. And a change in the political wind can reverse sensible changes: New York's new mayor is undoing many of the reforms that the ex-mayor implemented. Politicians can also sometimes treat the existence of innovation centres as a justification for carrying on with business as usual everywhere. What is more, **such centres spend a lot of time putting sticking plasters on cancers.** Hence, choice B is the most correct answer. The remaining choices do not apply. Choice (B)

**4.** Option D: Britain's Behavioural Insights Team, was the world's first government outfit dedicated to applying the insights of behavioural economics to public policy. Its mission was to point out the "small details" of policy that can have big consequences. It persuaded HM Revenue and Customs, Britain's tax collection agency, to **tweak** the words of a routine letter to say that most people in the recipient's local area had already paid their taxes. As a result, payment rates increased by 5 percentage points. .... Since this para talks about **the application of the insights of group behavioral economics to public policy**, we can infer that choice D is the correct answer.
Option A: Choice A is out of context.
Option B: While choice B may be relevant, it does not refer to the group behavioural aspect that the revised letter had included.
Option C: Choice C merely repeats an idea para from 3 but is not the reason for the question. The mission of Britain's Behavioural Insights Team was to point out the "small details" of policy that can have big consequences. Choice (D)

**Solutions for questions 5 to 8:**

**Number of words and Explanatory notes for RC:**

Number of words: 440

**5.** The second para mentions that animals communicate through gestures. Animals don't know about languages, gestures are insufficient for communication. "Language is considered the jewel in the crown of human superiority over other animals .... None can use these displays to convey a sentence like "I kicked the ball"." So the key idea in the second para is that humans have verbal language, something that other species don't.
The third para talks about the possible origin of human language.
So, there is a connection between the 2 paras and choice A is the correct answer.
Option B: The conclusion of the second para is that other animals cannot use language to communicate. The third para does not conclusively explain why other animals cannot use language. It only provides a probable explanation for how human language evolved. Hence, this is not the correct answer.
Option C: The second para and third para are along the same lines and there is no contradiction between the two. Hence, this is incorrect.
Option D: The second para provides the view that language is the domain of humans. The author does not refute this view. Hence, this option is incorrect. Choice (A)

**6.** Option A: Choice A has been stated in the first two sentences of para 2. Language is generally considered the jewel in the crown of human superiority ...... no other animal's communication comes close to it. Choice A is true and is not the answer.
Option B: Choice B has also been stated in the last sentence of para 2. But none can use these displays to convey a simple sentence such as "I kicked the ball". Hence, choice B is also true and is not the answer.
Option C: Choice C is also provided in the second para – see the last sentence. Hence, choice C is true and is not the answer.
Option D: Through clicks, hoots, barks, and bleats, some animals can inform each other of how they are feeling. But the contrast between language and gestures in choice D cannot be ascertained from the second para. Hence, choice D is the answer. Choice (D)

**7.** Option A: There is an overlap between parts of the mirror-neuron system and two areas in the brain's left cerebral cortex that are associated in humans with language production – Broca's and Wernicke's areas. This makes choice A the correct answer.
Option B: Choice B is not true. There is an overlap between the mirror neuron system and two areas in the brain's left cerebral cortex in humans.
Option C: Mirror neurons also seem to be involved with other actions. One is pointing (in children especially). The second part of choice C has not been explicitly stated in the passage.
Option D: Choice D is a distortion. Corballis assigns a central role to the brain's mirror neurons, which seem to echo action in observation (in monkeys, mirror neurons fire when they reach to grasp objects, and when they observe **another animal** doing the same). Choice (A)

**8.** Option A: The primatologist and ethologist Frans de Waal will find Corballis' assessment of primate vocalizations harsh. But choice A does not explain the complete essence of the quoted sentence in para 6. Hence choice A is not the answer.
Option B: Choice B is not the correct implication of the quoted sentence in the question. The author is not judgemental about Corballis' analysis of the origins of language and does not term the same unacceptable.
Option C: Many including the primatologist and ethologist Frans de Waal will find Corballis' assessment of primate vocalizations harsh. Hence choice C can be inferred.
Option D: Humans are on top in terms of communication because of language. The first part of choice D provides a reason for Corballis' assessment of primate vocalization. ... Corballis counters that primate vocalizations, unlike gestures, seem barely under voluntary control. He contends that they emerge like nervous tics, more closely connected to expression of emotion than to a deliberate exchange of information. However, this does not talk about the opinions of other primatologists, which is mentioned in the given statement. Hence, choice D is not an essence of the sentence given in the question. Choice (C)

**Solutions for questions 9 to 12:**

**Number of words and Explanatory notes for RC:**

Number of words: 541

**9.** Option A: Consider this statement: *'If future studies reveal a receptor present on this subset of neurons – but absent from most other cells – researchers could try to target it with drugs to selectively bump up activity.'* It shows that the author is looking forward to future studies that can reveal such receptors. Such a receptor will help because then these neurons can specifically be targeted using drugs, to see how their activity, appetites and weight gain are closely related. Hence, this finding will add depth to this research. Option A is not the answer.
Option B: While there is a correlation, we haven't quite determined whether there is causation as can be understood from *'It also doesn't prove that the change in brain activity itself caused the animals to overeat and gain weight.'* So, a study that proves that the activity of the neurons increases to ***prevent overeating in mice*** will close the open thread in this research – as raised by the author in the last para. Hence, Option B is not the answer.
Option C: Consider the following lines: *'But the study doesn't distinguish **which changed the neural activity: a feature of the diet or the weight gain itself.** It also doesn't prove that the change in brain activity itself caused the animals to overeat and gain weight'.* So, this research will answer the unanswered question raised by the author as to what causes the reduced functioning of the neurons. Hence, Option C is not the answer.
Option D: That there is a correlation has been established in the passage especially with the use of calcium imaging. This can be understood from the passage: *'If a lean mouse had just eaten, the neurons were more active in response to the sugar than if it had just been fasting. The cells seemed to act as a brake, signalling, "That's enough!"'* So, in the absence of obesity, when there is overeating, the neurons were more active. Hence, the correlation is already established. So, this will not add more depth to the research. Option D is the answer. Choice (D)

**10.** Option A: The author didn't really mean that the mice will be given electrical zaps during this research. What the author meant was that since mice don't enjoy electrical zaps, it was inferred that over-stimulating those neurons might be unpleasant for them as well, and hence, it is important not to overdo it. Option A is not the answer.
Option B: This is talking about the purpose of the future research – to find receptors and selectively target with drugs to study how those neurons function during overeating and obesity. It doesn't explain why the author emphasizes on a delicate tap instead of slamming the brakes with full force. Hence, Option B is not the answer.
Option C: This is the reason why the delicate tap is important. Overstimulation of neurons is not something the mice will like, as understood from how they react to electrical zaps given in that area. Hence, Option C is the answer.
Option D: This has got nothing to do with stimulation of neurons. Only after such receptors are discovered (receptors present on neurons being studied and nowhere else) can this experiment of selective targeting with drugs be even carried out. During that experiment, one of the things the author warns against is overstimulation of neurons. Hence, this option is easy to eliminate. Option D is not the answer. Choice (C)

**11.** Option A: The experiment will prove that the mice have been driven by taste towards overeating and not by the glutamatergic neurons. Hence, it is established glutamatergic neurons don't control appetites, taste does. Option A is not the answer.
Option B: Since the mice were overeating when the diet was tasty and came back to normal when the diet was bland, it can be established that taste does guide the appetites of the mice more than the internal mechanisms to prevent overeating. Hence, Option B is not the answer.
Option C: When they start eating high-fat mouse diets and their weight increases and then upon reverting their weight is lowered too, then it can be established that high-fat mouse diets had led to weight gain in case of the mice. Hence, Option C is not the answer.

Option D: If the mice overeat in one case and don't in another and get obese in one case and lose weight in another, and the activity of neurons goes up and down too, we can establish that overeating and neuron activity are correlated. But correlation is not causation, and hence, we cannot establish that the overeating affects neuron activity (this experiment doesn't establish that causation). Hence, Option D is the answer. Choice (D)

**12.** Consider the para: *'Changing an animal's diet "is a subtle manipulation" ... **The fact that you can go in and see different properties of these cells** is an amazing finding. ... wanted to understand the brain changes that accompany obesity. They were particularly interested in an area at the bottom of the brain known to regulate feeding, called the lateral hypothalamus. ... By **comparing gene expression between obese mice on a high-fat diet and control animals on a standard one**, the researchers identified a subset of neurons that changed most dramatically with the obesity-inducing diet.'*
Option A: The speed at which the diet change has been done as part of the experiment hasn't been explained in the passage. So, we cannot really infer that the author meant slow dietary changes when the term 'subtle manipulation' was used. Option A is not the answer.
Option B: The author calls it subtle manipulation because by simply changing the diet it becomes possible to study something very important – neurons which change dramatically because of the diet and ***what their possible functions are***. Hence, it is manipulating something externally to learn something much deeper and critical. Hence, Option B is the answer.
Option C: The author didn't compare the magnanimity of the diet changes with the reactions it triggers. Also, the author doesn't assert anywhere in the passage that ***even the smallest dietary change*** can produce ***huge changes in the brain***. That is entirely inaccurate. Hence, Option C is not the answer.
Option D: While diet is an easy thing to change, we don't know if there are other possibilities in order to study the neurons – since the study itself is about the connection between diet and the neurons that regulate eating, seemingly. The author's assertion was about how easy it is to change diet to study the neurons. It was NOT how easy it is to change the diet compared to other aspects of an animal's routine. Hence, Option D is not the answer. Choice (B)

**Solutions for questions 13 to 16:**

**Number of words and Explanatory notes for RC:**

Number of words: 530

**13.** The radical steps can be understood from: *'is **selling its unit** that makes **accelerator pedals, keypads and auto-body parts** after getting squeezed by Germany's carmakers. Instead, it'll **focus on public transport** and -- in a completely new endeavour -- **making lasers**.'*
Option A: This has been mentioned in the lines above, that Huebner will now focus on making lasers. Hence, Option A is not the answer.
Option B: This is true as well as understood from *'**The shift to new markets** was made because big manufacturers are mistreating suppliers by "forcing them into ever lower prices...'* So, it can be understood that the radical step involves moving away from being suppliers to big car manufacturers to other markets. Hence, Option B is not the answer.
Option C: We cannot quite infer that Huebner is closing down the rubber and plastics business. All we know is that Huebner is shifting from one set of products to a different set. Hence, Option C is the answer.
Option D: This has been mentioned in the above lines – that Huebner is selling its unit that makes auto-body parts and instead gearing up to focus on public transportation and lasers. The lasers are useful in biomedical research as understood from *'... The company's products are now used for holographic films for augmented reality glasses and biomedical research'.* Hence, Option D is not the answer. Choice (C)

**14.** The problems of the SMEs can be understood from the following lines: *'Often family-owned, they don't have the cash buffers to sit out a downturn. Nor are they well-positioned to address structural challenges like a dramatic shift in the underpinnings of modern manufacturing. The SMEs are more-or-less sandwiched" between big companies and their customers...'*
Option A: SMEs do not have the cash buffers to wait patiently for the downturn (or impending recession) to blow over, as per the lines above. That is the reason why they are particularly in a precarious situation. Hence, Option A is not the answer.
Option B: There is a dramatic shift in the underpinnings of modern manufacturing. The SMEs don't have to upgrade to any new 'technologies.' Rather, the author means that the SMEs are not positioned to address the challenges arising because of these shifts. From this we cannot conclude that the SMEs cannot modify their manufacturing techniques. Hence, Option B is the answer.
Option C: The clients of the SMEs are mostly the big car manufacturers 'mistreating suppliers by "forcing them into ever lower prices"' as understood from the passage. This is indeed a reason for their getting badly affected by the situation. Hence, Option C is not the answer.
Option D: The SMEs are getting sandwiched because of big companies cutting down on costs and pinching them for every penny. The prices being slashed for the suppliers means that the SMEs cannot run their businesses sustainably. So, this is not the answer. Choice (B)

**15.** Consider the following sentences: *'The ultimate challenge facing German manufacturing is the transition to so-called Industry 4.0, ... While Germany originated the term, the country is ill-equipped to deal with the transition, with deficiencies in everything from software training to Internet infrastructure...'*
Option A: The country is ill-equipped to deal with the transition to Industry 4.0 because of deficiencies in everything – from software training to the infrastructure for internet as understood from the lines - *While Germany originated the term, the country is ill-equipped to deal with the transition, with deficiencies in everything from software training to Internet infrastructure.* Hence, Option A definitely strengthens the argument that the transition is a huge challenge. Option A is the answer.
Option B: While it is true that the Mittelstand isn't getting much government support and is dealing with its own issues, there is no link between this and Germany's "transition" to Industry 4.0 which is at a much bigger scale. We need to understand that the Mittelstand is just a part of the manufacturing industry with its own problems, and those problems do not justify the argument about the transition. Hence, Option B is not the answer.
Option C: While companies are going through a high-risk shift to greener products and smarter industries, we cannot quite infer the trigger for this shift or the driver. We cannot quite understand that it is because of the demand for 'greener products.' It could be a change that the industry has effected on its own and not necessarily reacting to a demand. And there is no way to link this to the argument about challenges faced by Germany in 'transitioning'. Hence, Option C is not the answer.
Option D: Once again, it is a problem faced by Mittelstand and not one of the challenges faced by German manufacturing industry as a whole (since there are the larger players as well). The argument is related to the industry as a whole and the challenge it faces in transitioning to Industry 4.0. So, this option doesn't strengthen that argument. Option D is not the answer. Choice (A)

**16.** Option A: These are going for a spin because of the storm. Therefore, these cannot be the storm as understood from *'to cope with **a perfect storm that has sent traditional economic pillars** into a tailspin and put the country on the verge of recession'.* Hence, Option A is not the answer.
Option B: The recession has not yet come. The recession is not the cause for the trouble (the storm). Rather, it is a possible consequence. Hence, Option B is not the answer.
Option C: The perfect storm is most possibly a metaphor for the troubles faced by Mittelstand and Germany's manufacturing industry. We can understand that these storms are of two types: dramatic shifts in the underpinnings of modern manufacturing and impact of geopolitical shifts. Hence, this option rightly indicates what is the storm that is having such an adverse impact on the German economy. Hence, Option C is the answer.
Option D: It is not competition alone or predominantly that is the cause of the lowering the profit forecasts of the German giants. This can be understood from: *'**With day-to-day business roiled by geopolitical power plays**, German giants like Siemens AG, BASF SE and Daimler AG have been forced to lower profit forecasts and seek ways to cut costs.'* Hence, Option D is not the answer. Choice (C)

**Solution for question 17:**

**17.** Sentence 1 is an independent sentence that connects machine translation with the idea of understanding.
Sentence 2 is a question. There are a couple of things to notice here. The 'in fact' in this sentence is a clue that this is not an independent question, but a counter argument to something. Now this question is about whether understanding is needed for translation. That means the preceding sentences have to propagate the idea that there is a possible connection between translation and understanding.
Sentence 3 starts with a negative connector called 'Instead' which obviously breaks the flow of previous thoughts and goes in a counter direction. The other clue in this sentence is the usage of the definite article – 'the'. 'The field' signifies that some field has already been discussed in the para.
Sentence 4 is an independent sentence that introduces the idea of translation and also the negative aspect of how 'understanding' is missing. If we look closely Sentences 1 and 3 follow the same direction as 4 in establishing that understanding is not part of the process in translation. But, since 3 is talking about one particular field and how it 'has tried to decode', we can establish that 3 follows 1, which in turn follows 4. The 'instead' in 3 also fits because it talks about how machine translation '***actually***' works, since 1 talks about how it ***doesn't work.*** That leaves us with 2 which is a question, almost countering the idea that understanding is important for translation. It is also important to observe that another common order worth pursuing is 4213. However, in that case, for 1 to follow 2, as an example of the question raised by 2, 1 must begin with a positive connector, preferable, 'for example'. That is what makes 4132 a better, more coherent order. Ans: (4132)

**Solution for question 18:**

**18.** The para is divided into three ideas: 1. Postmodernism has bettered the way photographs can be analysed. 2. Photographs are now considered as valid historical evidence. 3. The information from photographs can be studied for context to help scholars.
Option A: It is not postmodernism that is directly helping scholarly research. The information that can be gleaned from the photographs can help scholars based on the refined notions introduced by postmodernism. Also, this summary misses out on the second idea mentioned above which is quite critical to the essence of the para. Option A is not the answer.
Option B: Scholarly research will benefit from the *rise in credibility of photographs* and the enhancement brought about by postmodernism in the *way photographs are analysed* for context. This option mentions all the three ideas of the para. That scholarly research will benefit because of two things – the new status of photographs as historically credible, as evidence for history, and the new way of looking at photographs ushered in by postmodernism. Hence, Option B is the answer.
Option C: There is no historic information being provided by photographs. Historic as opposed to historical means it is happening for the first time. Also, the way of looking at photographs has been refined. The information itself has not been refined by postmodernism. Hence, Option C is easy to eliminate.
Option D: The use of photographs for historical information is not thanks to postmodernism. Postmodernism is responsible for the way in which information can be gleaned from photographs. This is misrepresentation of information and can be easily eliminated. Option D is not the answer. Choice (B)

**Solutions for question 19:**

**19.** It cannot be (1) because the given sentence is downstream to the sentences following (1). We are talking about people trying to churn antimetaboles without understanding that there needs to be a logical theme. The ideas in the first two sentences are still bigger than the idea of the actual construction of the sentence (which is what the bold sentence talks about).
It cannot be (2) – that is obvious to spot – because the sentence following (1) says 'they' pointing to 'people' in the previous sentence, and, hence, we cannot sneak in a sentence about the construction of antimetaboles.
It cannot be (3) because we are still talking about 'antimetaboles' as the bigger idea and not the construction of the sentence itself. And we are still talking about 'appeal to reason'.
It is apt at (4) because it is sandwiched between the reason for popularity of antimetabole (which is carried forward in the bold sentence – that people can guess the second part) and an example. We give the example, after the complete explanation. Hence, the answer is (4), i.e., option D. Choice (D)

**Solutions for questions 20 and 21:**

**20.** Sentence 1 starts with a connector 'or rather' with a personal pronoun 'they' pointing to a group of people.
Sentence 2 talks about 'the format' probably discussing some previously defined format. (Please note that words like 'though' and 'despite' do not count as connectors).
Sentence 3 is an independent sentence that starts a discussion, about how some Korean high-school students would want to be a comic artist.
Sentence 4 is an independent sentence that describes webtoons and their format. From this we can understand that webtoons are the format being discussed in 2.
Being a comic artist in Sentence 3 is the biggest of all ideas and is also a precursor to the first sentence where 'they' are probably Korean high-school students. So, 31 is a block.
Similarly, 'also' in Sentence 2 is the other clue. The format is for cell phones. The format is ***also*** a connector to the past. So, 42 is a block. 4 is a continuation of the 31 block in explaining the webtoon format. So, 3142. Ans: (3142)

**21.** Sentence 1 starts with a negative connector – 'but' followed by something that suggests a protest/fight – call to arms.
Sentence 2 talks about how imagining something bad happening to children is just an act of imagination – contrasting with how it should be more. That is what justifies Sentence 1 – should not just be imagination, but a call of arms.
Sentence 3 talks about our response to scenarios – instigating/participating/ignoring (turning a blind eye) and how we will be remembered for those responses.
Sentence 4 asks a question – about the consequences of negative acts towards the smallest humans – connecting to Sentence 2 – children.
Sentence 3 and 4 raise the question about how we would respond to something bad. 4 follows 3 because 3 is more generic/upstream while 4 talks specifically about children.
21 is a block because of the connector in 1. Once, the discussion moves to call of arms, it can't lead to the questions in 3 and 4. Also, 4 raises the question, while 2 talks about empathy by bringing the action closer from any children, to 'one's own children'. So, 3421. Ans: (3421)

**Solutions for question 22:**

**22.** It cannot be Option (1) because the subsequent sentence starts with 'That matters', which means we are expecting something definitive in the previous line. The discussion of Kivas can very well lead to the line that starts with 'that matters'. Also, the line 'that matters' is still building towards the important idea - It is not just what you think and believe that is important in politics, but also that others know that you know. So, the given bold line is downstream to this.
Option 2 is the apt place for the bold line, since it is an elaboration of explanation of the previous statement 'I need to know that you know I'm willing.' Also, this elaboration must come before we give it a name – common knowledge.

It can't be Option 3 because the two lines on either side of Option 3 are connected. 'This' (whatever has been mentioned before) is called common knowledge and this is important for costly cooperation. There is nothing specifically pointed to in the given bold line that can be said to be what is important for costly cooperation.
It cannot be Option 4 at the end of the para because the given bold line is upstream to the discussion of what common knowledge is. By the time we come here, we are already talking about the practical application of common knowledge. The given bold line is still building towards the idea of what common knowledge means. Choice (B)

**Solutions for questions 23 and 24:**

**23.** Option A: The focus of option A is about heart diseases, whereas the para talks about being healthier and living longer. Hence, this is not the answer.
Option B: The para does not provide enough data to conclude that "most heart diseases are due to stress and most of the stress is due to problems between spouses". Hence, this is also incorrect.
Option C: This option mentions the happiness of marriage and link between marital stress and health. It also talks about sharing interests. Hence, this is the correct answer.
Option D: The first part of option D is inverted. The para mentions that couples with good marital relationships share interests. The option says that couple with shared interests have happy marriages. Also, the second part of option D is not correct. It is only likely that heart diseases can develop due to marital stress. The option mentions it as a fact and not as a possibility. Hence, option D is eliminated. Only choice (C) gives the complete picture. Choice (C)

**24.** Option A: This option presents all the three aspects that the para is trying to convey. Firstly, high profile people are entering into the climate change debate, and they are adding their voice for action against climate change. Secondly, there are a bunch of rational individuals who believe nothing of note can be done now. Thirdly, the scepticism has been deepened by these rational individuals. Hence, Option A represents the essence of the passage.
Option B: A lot of well-known people are acknowledging the need for action. They have not turned sceptical. So, this option can be easily eliminated. Option B is not the answer.
Option C: The debate has had two additions – high profile people on the pro-climate action side and rational analytical ones on the sceptical end. The option assumes it is high-profile individuals on either side. That is incorrect. Option C is not the answer.
Option D: Heads of state, regulators and CEOs have not taken the place of academicians and activists. They have merely supplemented the latter. Also, this option omits the more important aspect about rational people joining the scepticism bandwagon believing that it is too late for climate change action. Hence, Option D is not the answer. Choice (A)

| *Difficulty level wise summary - Section I* | |
|---|---|
| Level of Difficulty | Questions |
| *Very Easy* | – |
| *Easy* | – |
| *Medium* | 1, 2, 4, 6, 7, 8, 12, 13, 14, 15, 20, 23, 24 |
| *Difficult* | 3, 5, 9, 10, 16, 17, 19, 22 |
| *Very Difficult* | 11, 18, 21 |

## SECTION – II

**Solutions for questions 1 to 5:**

Let M, U and K represent the panel members. From the scatter chart, we can find the rating given by the members (but we cannot determine to which candidate the rating was given). Rewriting this information,

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Mishra** | 1 | 1 | 4 | 5 | 6 | 7 | 7 | 8 | 8 | 9 |
| **Uday** | 2 | 2 | 3 | 4 | 4 | 6 | 6 | 7 | 9 | 9 |
| **Kumar** | 1 | 2 | 3 | 3 | 5 | 5 | 6 | 6 | 6 | 8 |

The cells in grey represent the ratings already assigned to candidates (from the table in the question).

A received a final score of 4. Hence, the sum of the ratings given by Mishra, Uday and Kumar combined must be 4 × 3 = 12. M and U combined must have given a rating of 6. Mishra and Uday can give a combined rating of 6 only if Mishra gave 4 and Uday gave 2.

For B, Mishra and Uday together should have given 9. This is possible if Mishra and Uday gave 5 and 4 OR 7 and 2 respectively. But Uday has given only two ratings of 2 – to J and A. Hence, Mishra and Uday gave 5 and 4 respectively.

For C, Mishra and Kumar together must have given 14. Possibilities are 8 and 6 OR 9 and 5.

For E, Uday must have given 3. For F, the only possibility for Mishra and Kumar are 7 and 5. Now, for C, Mishra and Kumar could not have given 9 and 5. Hence, they must have given 8 and 6.

For G, Mishra and Uday must have given a total of 7. Hence, 1 and 6 is the only possibility.
For J, Mishra and Kumar must have given 1 and 6 respectively.

Since the only rating left for Kumar is 2, Kumar must have given that rating to H. Mishra and Uday must have given 9 and 4 respectively.

Since Mishra has only rating left, he must have given 7 to I. Uday could have given 7 and 9 to either of I and D. If Uday gives 9 to D, the final score of the candidate will exceed 8. Hence, Uday gave 9 to I and 7 to D.
The final table of ratings is given below:

| | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|
| **Mishra** | 4 | 5 | 8 | 8 | 6 | 7 | 1 | 9 | 7 | 1 |
| **Uday** | 2 | 4 | 6 | 7 | 3 | 9 | 6 | 4 | 9 | 2 |
| **Kumar** | 6 | 1 | 6 | 8 | 3 | 5 | 5 | 2 | 3 | 6 |
| **Final Score** | 4 | 3.33 | 6.67 | 7.67 | 4 | 7 | 4 | 5 | 6.33 | 3 |

**1.** Uday gave a rating of 9 to I. Choice (D)

**2.** Three candidates – C, D and E – received the same rating from two members. Choice (B)

**3.** For 5 candidates, A, B, E, G and J, the rating given by any member was not greater than 7. Choice (D)

**4.** The highest final score received by a candidate was 7.67. Choice (C)

**5.** Mishra gave a rating of 1 to both G and J. Choice (B)

**Solutions for questions 6 to 10:**

Let the total number of potatoes that they purchased be a. From (i), the number of tomatoes that Gaurav purchased is also a.

From (v), Amit purchased four cauliflowers; Kiran purchased five tomatoes; Gaurav purchased six onions.

From (iii), Amit must have purchased six tomatoes. Since only 3 persons purchased tomatoes, the total number of tomatoes that they purchased must be a + 5 + 6 = a + 11.

From (iv), the number of cauliflowers they purchased must be a – 4 and the number of onions they purchased must be a – 29.

From (ii), total number of vegetables that Pranav purchased must be a – 4.

Also, Pranav did not purchase any tomatoes (since three persons, Amit, Kiran and Gaurav, purchased tomatoes)

Pranav could not have purchased any potatoes either. This is because if he had purchased potatoes, it must have been a potatoes (since only one person purchased potatoes). But the total number of vegetables that Pranav purchased is only a – 4. Hence, Pranav could have purchased either cauliflowers and onions or only onions. If Pranav purchased only Onions, he must have purchased a – 4 onions. But the total number of onions that the four of them purchased is only a – 29, which is less than a – 4. Hence, Pranav must have purchased both cauliflowers and onions.

Since the total number of cauliflowers purchased is a – 4 and Amit purchased four cauliflowers, Pranav must have purchased a – 8 cauliflowers. This implies that he must have purchased 4 onions.

The table below provides the number of vegetables that each person purchased. An 'X' indicates that the person did not purchase that type of vegetable.

| | Potatoes | Cauliflower | Tomatoes | Onions | TOTAL |
|---|---|---|---|---|---|
| Amit | | 4 | 6 | | |
| Kiran | | X | 5 | | |
| Pranav | X | a – 8 | X | 4 | a – 4 |
| Gaurav | | X | a | 6 | |
| TOTAL | a | a – 4 | a + 11 | a – 29 | |

**6.** The minimum number of onions that they could have purchased is 6 + 4 + 1 + 1 = 12.
Hence, a – 29 ≥ 12 ⇒ a ≥ 41
The minimum number of tomatoes that they could have purchased = a + 11 = 52. Ans: (52)

**7.** The number of onions that Pranav purchased is 4. This lies between 3 and 6. Choice (C)

**8.** The only person who could have purchased all the four types of vegetables is Amit.
Hence, Amit must have purchased a potatoes. Kiran and Gaurav could not have purchased any potatoes.
Kiran and Amit must have purchased a total of a – 39 onions.
If Amit purchased a – 40 onions and Kiran purchase 1 onion, then the total number of vegetables purchased by Amit will be 2a – 30, by

Kiran will be 6, by Pranav will be a – 4, by Gaurav will be a + 6.
If Amit purchased 1 onion and Kiran purchased a – 40 onions, then the total number of vegetables purchased by Amit will be a + 11, by Kiran will be a – 35, by Pranav will be a – 4, by Gaurav will be a + 6.
Since a must be greater than 41, in both the cases, Amit would have purchased the highest number of vegetables. Choice (A)

**9.** If Amit purchased a total of 2a – 30 vegetables, then the minimum number of vegetables = 2 × 41 – 30 = 52
If Amit purchased a total of a + 11 vegetables, then the minimum number of vegetables = 41 + 11 = 52
Hence, the minimum number of vegetables that Amit purchased = 52. Ans: (52)

**10.** Pranav purchased a total of a – 4 vegetables. Gaurav purchased a + 6 tomatoes and onions. Hence, the total number of vegetables that Gaurav purchased must be more than a – 4.
Pranav could not have purchased the highest number of vegetables. Choice (C)

**Solutions for questions 11 to 15:**

Let 1 to 9 represent the positions of the chairs in the clockwise direction. Let the person who arrived first sit at 1. The person who arrived second will sit at 6. The person who arrived third will sit at 2. Similarly, the persons who arrived fourth, fifth, sixth, seventh, eighth and ninth will sit at 7, 3, 8, 4, 9 and 5 respectively. We can take the order in which they arrived to be a proxy for their positions around the table.

From (i), B must be at 3 or 8, i.e., B must have been the fifth to arrive or the sixth to arrive.

From (iii), H arrived after D and H was sitting two places to the right of D.

D and H could have arrived 1st and 6th OR 3rd and 8th OR 4th and 9th OR 2nd and 7th. From (vi), D was not the second to arrive and H was not the last to arrive. Hence, they could not have arrived 4th and 9th OR 2nd and 7th.

Let D and H be the 1st and 6th to arrive. B must be the 5th to arrive.

From (iv) and (vi), G and I can be the 3rd and 6th to arrive OR they can be the 2nd and 5th to arrive (in each case, two persons arrived between them and two persons sitting between them). But neither of the two cases are possible since H was the sixth to arrive and B was the fifth to arrive. Hence, this case is not possible.

Hence, D and H must be the 3rd and 8th to arrive. G and I cannot be the 3rd and 6th to arrive. Hence, they must be the 2nd and 5th to arrive. B must be the sixth to arrive.

From (ii), C cannot be the 1st to arrive, since he is not adjacent to H. C must be the 4th to arrive, since he arrived before I (who was the 5th to arrive).

From (i), E cannot be the 1st to arrive, since he is not adjacent to H. E must be the 7th to arrive. A is not the last to arrive. Hence, A must be the first to arrive and F must be the last to arrive.

The following table provides the positions of the nine persons (from 1 to 9) around the table and the order in which they arrived:

| Position | Order | Person |
|---|---|---|
| 1 | 1 | A |
| 2 | 3 | D |
| 3 | 5 | I |
| 4 | 7 | E |
| 5 | 9 | F |
| 6 | 2 | G |
| 7 | 4 | C |
| 8 | 6 | B |
| 9 | 8 | H |

**11.** Three persons arrived before C did. Ans: (3)

**12.** H was the 8th person to arrive. Two seats were empty just before H arrived at the conference. Ans: (2)

**13.** Among the two persons who were sitting adjacent to D, A arrived before D and hence, he would have been sitting in a chair adjacent to the one in which D sat by the time D arrived. Choice (C)

**14.** G was the second person to arrive. Among the given options, C was sitting adjacent to G. Choice (D)

**15.** A was the first person to arrive at the conference. Choice (A)

**Solutions for questions 16 to 20:**

Let n be the total number of seats and m be the total revenue if all the seats were filled. The following table provides the number of seats, revenue and price for each type of seat:

| Type of Seat | Number of Seats | Revenue | Price of each seat |
|---|---|---|---|
| A | 0.2*n* | 0.24*m* | $\frac{1.2m}{n}$ |
| B | 0.05*n* | 0.12*m* | $\frac{2.4m}{n}$ |
| C | 0.25*n* | 0.2*m* | $\frac{0.8m}{n}$ |
| D | 0.15*n* | 0.3*m* | $\frac{2m}{n}$ |
| E | 0.35*n* | 0.14*m* | $\frac{0.4m}{n}$ |

**16.** The price of seat type B is the highest. Choice (B)

**17.** Given that the revenue from each type of seat was the same, let the revenue from each type of seat be k. k can take a maximum value of 0.12m (since the revenue from seats of type B can never be more than 0.12m). Hence, if all the seats of a particular type were filled, then the type of seat B would have been sold out. Choice (A)

**18.** Given that the price of seat type C is ₹80. Hence, m/n = 100
The price of the seats A, B, D and E will be 120, 240, 200 and 40 respectively.
Since the revenue is the same, and we need to minimize the number of seats, we also need to minimize the revenue from each type of seat. Calculating the LCM of 40, 80, 120, 200 and 240 will give the minimum possible revenue from each type of seat.
The minimum possible revenue from each type of seat = 1200
Number of seats of type A that were filled = 1200/120 = 10
Number of seats of type B that were filled = 1200/240 = 5
Number of seats of type C that were filled = 1200/80 = 15
Number of seats of type D that were filled = 1200/200 = 6
Number of seats of type E that were filled = 1200/40 = 30
Minimum number of seats that would have been filled = 10 + 5 + 15 + 6 + 30 = 66 Ans: (66)

**19.** If 60% of each type of seat was filled, the revenue from seats of type A will vary between 0.24m × 0.6 = 0.144m to 0.24m
Similarly, we can find the variation in revenue for each seat type. The following table presents this range:

| Type of Seat | Range of Revenue |
|---|---|
| A | 0.144m to 0.24m |
| B | 0.072m to 0.12m |
| C | 0.12m to 0.2m |
| D | 0.18m to 0.3m |
| E | 0.084m to 0.14m |

By observation, we can see that at most three types of seat would have given the same revenue (C, B and E for 0.12m OR A, C and D for 0.18m to 0.2m). Choice (C)

**20.** From the above question, we can see that for at least 60% of seats filled, revenue from three types of seats are the same. This needs to decrease so that the revenues from four types of seats overlap. By observation, we can see that if revenue of 0.144m from seat type A can decrease to 0.12m, this will make the revenues from A, C, B and E the same (revenue from each type of seat will be 0.12m).
Since the revenue from seat type A if all seats are filled is 0.24m, if 50% of seats are filled, then the revenue will become 0.12m. Hence, the maximum possible value of p is 50.
We can also notice that for p = 50, no other combination of four types of seats having the same revenue is possible. Ans: (50)

| *Difficulty level wise summary - Section II* | |
|---|---|
| Level of Difficulty | Questions |
| *Very Easy* | |
| *Easy* | 16 |
| *Medium* | 11, 12, 13, 14, 15, 17 |
| *Difficult* | 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 18, 19, 20 |
| *Very Difficult* | |

## SECTION – III

**Solutions for questions 1 to 22:**

**1.** Number of non-empty subsets of any set having N elements = $2^N – 1$.
Number of non-empty subsets of S = $2^{400} – 1$
The non-empty subsets of S which have the product of their elements equal to on odd number have each of their elements equal to an odd number. These subsets of S are subsets of {1, 3, 5, ...., 399}.
There are $2^{200} – 1$ non-empty subsets of S which have the product of their elements equal to an odd number.
∴ The remaining $2^{400} – 1 – (2^{200} – 1)$,
i.e., $2^{400} – 2^{200}$, non-empty subsets of S have the product of their elements equal to an even number.
$2^{400} – 2^{200} = 2^{200}(2^{200} – 1)$ Choice (C)

**2.** The largest number which leaves remainders of 4, 6 and 2 when it divides 460, 690 and 990 respectively will be
HCF (460–4, 690–6, 990–2)
= HCF (456, 684, 988)
= HCF (684 – 456, 988 – 684)
= HCF (228, 304) = 76.

Hence, the required number = 76. Ans: (76)

**3.** Since there is no student who has passed in more than 5 subjects, the number of students who have passed in at least three subjects, (i.e., the required answer) = number of students who passed in *exactly 3 + exactly 4 + exactly 5* subjects.

Now, from the data given the number of students who passed in not more than 2 subjects is given as 9. This is simply the number of students who passed in *exactly 0 + exactly 1 + exactly 2* subjects.

Similarly, the number of students who passed in not more than 5 subjects is given as 22. This is simply the number of students who passed in *exactly 0 + exactly 1 + exactly 2 + exactly 3 + exactly 4 + exactly 5 subjects.*

Hence, the required answer = 22 – 9 = 13.

**Alternative Solution:**
The number of students who passed in at most 0 subjects is 3 i.e., the number of students who passed in exactly 0 subjects is 3.
The number of students who passed in at most 1 subject is 7, i.e. the number of students who passed in exactly 1 subject is 7 – 3 or 4.
We can proceed along similar lines and work out the number of students who passed in exactly 2, 3, 4 and 5 subjects.
The data and calculations are tabulated below. The data appears in the second row.

| | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| No. of students who passed in exactly | 3 | 4 | 2 | 5 | 1 | 7 |
| No. of students who passed in at most | 3 | 7 | 9 | 14 | 15 | 22 |
| No. of students who passed in at least | 22 | 19 | 15 | 13 | 8 | 7 |

Each number in the first row (except the first) is the corresponding number in the second row minus the number to its left.
Each number in the third row (except the last) is the sum of all the numbers in the first row on top and to the right in the first row.
∴ The number of students who passed in at least 3 subjects is 13. Choice (B)

**4.** Let the height of the room be *h* ft.
Lateral surface area of the room = 2*h* (*l* + *b*)
= 2*h* (15 + 12)
= 54*h*

The cost of painting the four walls = (54*h*) (25)
It is given that (54*h*) (25) = 10800
⇒ *h* = 8 ft. Ans: (8)

**5.** The relative speed of the car with respect to the cart
= 44 – 8 = 36 km/hr
$= 36 \times \frac{5}{18} = 10$ m/s. Now, the car would be visible to the driver of the cart till the car is 80 m ahead of the cart.
Time for which the car is visible
$= \frac{80}{10} = 8$ seconds Ans: (8)

**6.** Let the two series be
$a_1, a_1r_1, a_1r_1^2 \ldots\ldots a_1r_1^{n-1}$
$a_2, a_2r_2, a_2r_2^2 \ldots\ldots a_2r_2^{n-1}$
Given $a_1r_1^3 = a_2r_2 \to (1)$
$a_1r_1^5 = a_2r_2^2 \to (2)$
(2) ÷ (1) gives $r_2 = r_1^2 \to (3)$
and (3) in (1) gives $a_2 = a_1r_1$ —— (4)
also given $a_2r_2^{n-1} = 64a_1r_1^{n-1} \to$ (5)
substituting $r_2 = r_1^2$ and $a_2 = a_1r_1$ in (5)
$r_1^n = 64 \to (6)$
Also given fifth term of first series = 8(second term)
$\Rightarrow a_1r_1^4 = 8a_1r_1 \quad \Rightarrow r_1^3 = 8 \quad \Rightarrow r_1 = 2 \to (7)$
Now $2^n = 64 \quad \Rightarrow n = 6$ Choice (B)

**7.** Clearly, the ages of A, B and C are in A.P., with A being the middle term. The average of B, A and C will be equal to A itself. Hence A's age = 19 and B's age = 19 – 5 = 14. Choice (A)

**8.** Let the cost price, the selling price and the marked price of the article be denoted by C, C + *d* and C + 2*d* respectively.

Percentage profit = $\frac{d}{C} \times 100$

Discount percentage = $\frac{d}{C+2d} \times 100$

Clearly, the percentage profit is more than the percentage discount. Thus, option (A) and option (B) are ruled out

Markup percentage = $\frac{2d}{C} \times 100$

Therefore, the mark up percentage is twice the percentage profit. Hence, option (C) is true.
Now the percentage profit when calculated on the selling price = $\frac{d}{C+d} \times 100$

Now, $\frac{d}{C+d} \times 100 < \frac{d}{C} \times 100$. Thus, option (D) is incorrect. Choice (C)

**9.** $\frac{1}{1!} \times \frac{2}{2!} \times \frac{3}{3!} \times \frac{4}{4!} \ldots\ldots \frac{n}{n!}$

$= 1 \times 1 \times \frac{1}{2!} \times \frac{1}{3!} \times \ldots\ldots \times \frac{1}{(n-1)!}$

$= \frac{1}{2! \times 3! \times 4! \ldots\ldots (n-1)!}$

$= \frac{1}{2^{n-2}\,3^{n-3}\,4^{n-4} \ldots\ldots (n-1)^1}$ Choice (C)

**10.** Let the number when expressed in base 6 be represented by $(a\,b\,c)_6$
It is given that $4\,(a\,b\,c)_6 = (c\,b\,a)_{11}$
⇒ 4 (36*a* + 6*b* + *c*) = (121*c* + 11*b* + *a*)
⇒ 143*a* + 13*b* = 117*c*
⇒ 11*a* + *b* = 9*c*
As the number is expressed in base 6, *b* can be any digit from 0 to 5.

If *b* = 0, then 9*c* = 11*a*
$\Rightarrow \frac{c}{a} = \frac{11}{9}$
This is not possible as *a* and *c* needs to be from 0 to 5.
If *b* = 1, 9*c* = 11*a* + 1
⇒ 9*c* = 9*a* + 2*a* + 1.
As L.H.S is divisible by 9, R.H.S must be divisible by 9 i.e., 2*a* + 1 must be divisible by 9.
∴ *a* = 4 and *c* = 5.
If *b* = 2, 9*c* = 11*a* + 2
⇒ 9*c* = 9*a* + 2*a* + 2
We do not get any possible value of *a* within the admissible range.
If *b* = 3, 9*c* = 9*a* + 2*a* + 3
As 2*a* + 3 is divisible by 9, we get *a* = 3 and *c* = 4.
If *b* = 4, 9*c* = 9*a* + 2*a* + 4.
Again 2*a* + 4 to be divisible by 9 does not give any value of *a* in the admissible range.
If *a* = 5, 9*c* = 9*a* + 2*a* + 5
As 2*a* + 5 is divisible by 9, we get *a* = 2 and *c* = 3.
Thus, the numbers satisfying the given criteria in base 6 are $(415)_6$ or $(334)_6$ or $(253)_6$.
Therefore, three possible values exist for N. Ans: (3)

**11.** As A and B are the sum of the first n and m terms of two geometric progressions and A = 3B, we have
$$\frac{1(2^{n+1}-1)}{2-1} = 3\left[\frac{1(4^{m+1}-1)}{4-1}\right]$$
$\Rightarrow 2^{n+1} – 1 = 4^{m+1} – 1 \Rightarrow 2^{n+1} = 4^{m+1}$
$\Rightarrow n + 1 = 2m + 2 \Rightarrow n – 2m = 1$

**Alternative Solution:**
By observation 1 + 2 = 3 × (1), i.e., *n* = 1 and *m* = 0
⇒ 1 – 2 (0) = 1. Ans: (1)

**12.** Let $\frac{x+2}{(x+1)(x+3)} = y$
$\therefore x + 2 = (x^2 + 4x + 3)\,y$
$x^2y + 4xy – x + 3y – 2 = 0$
$yx^2 + x(4y – 1) + 3y – 2 = 0$
as x is real, the discriminant ≥ 0
$(4y – 1)^2 – 4y\,(3y – 2) \geq 0$
$16y^2 – 8y + 1 – 12y^2 + 8y \geq 0$
$4y^2 + 1 \geq 0$
Now, $4y^2 + 1$ will be positive for any real value of *y*
Hence, range of *y* = (–∞, ∞). Choice (D)

**13.** Let AD = 2 ⇒ AB = 4 ⇒ FG = 1 and the altitudes of the triangles FGH, FGK and FGJ will be in the ratio 3 : 2 : 1 (Since FJ = JK = KM)
and these triangles are on the same base of FG = 1
⇒ altitude of GKJ, when JK is taken as base, = 1/3 of AD = 2/3
∴ Area of the shaded region = $\frac{1}{2} \times 1 \times \frac{2}{3} = \frac{1}{3}$
Area of ABCD = 4 x 2 = 8
∴ Required ratio = 1/3 : 8 i.e., 1 : 24. Choice (D)

**14.** The square of any natural number *n* leaves a remainder of 1 when divided by 24 if and only if *n* = 6*p* ± 1 (since *n* must be divisible by neither 2 nor 3) where *p* is a natural number.
∴ The two-digit numbers are of the form 6*p* ± 1.
There are 15 two-digit numbers of the form 6*p* + 1 and 15 two-digit numbers of the form 6*p* – 1.
∴ A total of 30 two-digit numbers satisfy the given condition. Ans: (30)

**15.** $5^{(\log_{25} 961)} = 5^{\left(\frac{1}{2}\log_5 961\right)} = 5^{\left(\log_5 961^{\frac{1}{2}}\right)}$
$= 5^{(\log_5 31)} = 31$

$7^{(\log_{\sqrt{7}} 29)} = 7^{(2\log_7 29)} = 7^{(\log_7 29^2)}$
$= 29^2 = 841$

$\therefore 13^{(\log_x 872)} = 31 + 841 = 872$
$\therefore x = 13$ Choice (D)

**16.** To obtain the required permutation we proceed as follows

| Permutation | Number of transpositions |
|---|---|
| abcdef | |
| bcdefa | 5 |
| cdefba | 4 |
| defcba | 3 |
| efdcba | 2 |
| fedcba | 1 |

∴ The total number of transpositions needed is 15. Choice (A)

**17.** Let the cost price of the article be ₹100
⇒ Its marked price = ₹160
Its selling price = $\frac{3}{4}$ (160) = ₹120
∴ Profit percentage on the article
$= 100 \times \frac{(120-100)}{100} = 20\%$ Choice (A)

**18.** Each time Turbo travels a one-way trip, i.e., from A to B or B to A, he and Black Shadow will meet exactly once, unless they meet at A or B (in which case, they will meet only once for two one-way trips by Turbo).
Each one-way trip by Turbo takes $\frac{7}{70} \times 60 =$ 6 minutes. Each one-way trip by Back Shadow takes $\frac{7}{20} \times 60$
= 21 minutes.

Now, let both of them meet (say) at A. Then an even multiple of 6 = an odd multiple of 21, which is not possible.
Now, let them meet at (say) B. Then, an odd multiple of 6 = an even multiple of 21, which is possible. Now, by some observation, we can figure out that they meet at B for the first time when Turbo travels from A to B four times and from B to A three times and Black Shadow travels B to A and then from A to B, once.
But they would not meet when Turbo travels from B to A, after which they would meet another six times by the time Black Shadow reaches B for the second time, at which instance they are back in their initial positions again.
So Turbo and Black Shadow meet thirteen times in $\frac{(14)(7)}{(70)} = \frac{7}{5}$ hours.

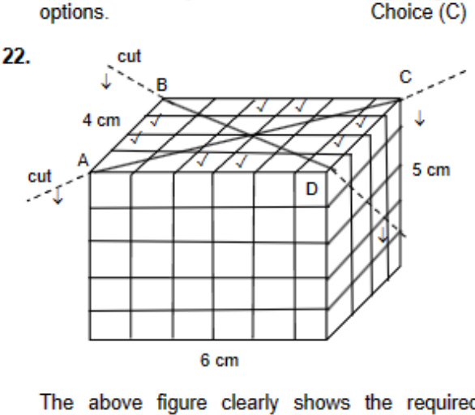
∴ In seven hours, they would meet $13 \times \frac{7}{\left(\frac{7}{5}\right)} = 65$ times. Ans: (65)

**19.** On the first day, the vendor can visit any one of the 10 villages. On the second day, the vendor can visit any one of the remaining 9 villages. On the third day, the vendor can visit any one of the remaining 8 villages.
From the fourth day onwards till the eleventh day, on each day he can visit any one of 8 of the 10 villages, i.e., excluding the villages he visited on the preceding two days.
Therefore, the number of ways in which the vendor can visit the villages in 11 days = $(10)(9)8^9$. Choice (B)

**20.** As the income of A is ₹15000, the expenditure of B is also ₹15000. Let the expenditure of A be *a* and the income of B be *b*, then (15000 – *a*) = 2(*b* – 15000)
⇒ 2*b* + *a* = ₹45000 Choice (C)

**21.** By substitution, we can observe that the points (0, 0), (–2, 0), (2, 0), (–1, –1) and (1, –1) satisfy only the relation y = | |x| – 1 | – 1 among the given options. Choice (C)

**22.**



The above figure clearly shows the required arrangement. The cuts are made along AC and BD, the two diagonals of the top face ABCD. The top layer of unit cubes, will have exactly eight cubes (indicated with a '√' mark) which are unaffected by the cuts. There will be five such layers in the cuboid. Hence 8 (5) or 40 unit cubes are unaffected by the cuts. Ans: (40)

| *Difficulty level wise summary - Section III* | |
|---|---|
| Level of Difficulty | Questions |
| *Very Easy* | 7, 17 |
| *Easy* | 2, 4, 5, 9, 13, 15, 20 |
| *Medium* | 3, 6, 8, 11, 12, 14, 16, 19, 21, 22 |
| *Difficult* | 1, 10, 18 |
| *Very Difficult* | – |

Done Tip: Scroll slowly for the best results